SMAI Spring 2026 – Final Examination
Answer Key & Grading Rubric

IIIT Hyderabad

General Grading Principles:

- Award partial credit generously when student shows correct conceptual understanding (for derivations and numerical work).
- For numerical answers, accept reasonable rounding (e.g., 0.61 for 0.6087).
- For derivations, award credit for correct setup and chain rule, even if final arithmetic has minor errors.
- Accept any valid notational convention (e.g., MSE with $\frac{1}{N}$ or $\frac{1}{2N}$ factor).

MCQ Grading Rule (Strict, applies to ALL multi-correct MCQs):

- **100% marks:** ALL correct options selected AND NO incorrect option selected.
- **50% marks:** At least half of the correct options selected AND NO incorrect option selected.
- **0 marks:** Any incorrect option selected (false positive), *OR* fewer than half of the correct options selected.

Note: "Half" means $\lceil n/2 \rceil$ where n is the number of correct options. E.g., for 3 correct options, at least 2 must be selected for 50% marks. For 2 correct options, at least 1. For a single-correct MCQ, only full marks (selecting it) or zero (not selecting / selecting wrong) is possible.

This rule supersedes any per-question MCQ partial-credit guidance below.

Part I: Objective Questions [60 marks]

Q1. IPL Wicket-Prediction Classifier [2+2+2 = 6 marks]

Question

Test set: 1000 balls (15 actual wickets). Model flags 20 as "wicket-likely" and gets 12 actual wickets correct.

Answer

(a) Confusion Matrix:

- TP = 12 (correctly flagged wickets)
- FP = 20 - 12 = **8** (flagged as wicket but actually no-wicket)
- FN = 15 - 12 = **3** (actual wickets missed)
- TN = 1000 - 15 - 8 = **977** (correctly classified no-wickets)

Verification: $12 + 8 + 3 + 977 = 1000$ ✓

(b) Precision $= \frac{TP}{TP+FP} = \frac{12}{20} = \mathbf{0.60}$; Recall $= \frac{TP}{TP+FN} = \frac{12}{15} = \mathbf{0.80}$

(c) Valid criticisms: **(i), (ii), (iii)**. Option (iv) is incorrect – confusion matrices ARE useful for imbalanced classes.

Grading Rubric

(a) [2 marks]: 0.5 mark for each of TP, FP, FN, TN. Award full marks if any 3 are correct (the 4th follows). Common error: swapping FP/FN – still award 1 mark.

(b) [2 marks]: 1 mark each for precision and recall. Accept fractional or decimal forms (12/20, 0.6, 60%). If TP/FP/FN values from (a) are wrong but ratios are correctly computed from those wrong values, award full credit (carry-forward).

(c) [2 marks]: Correct options: **(i), (ii), (iii)** (3 correct).

- **2 marks:** all of (i), (ii), (iii) selected AND (iv) NOT selected.
- **1 mark:** exactly 2 of (i), (ii), (iii) selected AND (iv) NOT selected.
- **0 marks:** (iv) selected (false positive), OR only 1 (or 0) of the correct options selected.

Q2. Linear Regression: Predicting Strike Rate [2+2+2 = 6 marks]

Answer

(a) FALSE. The closed-form $w^* = (X^\top X)^{-1} X^\top y$ requires $X^\top X$ to be *invertible* (non-singular). When $X^\top X$ is singular, the inverse does not exist and the solution is not unique (one may use pseudo-inverse).

(b) For $J(w) = \frac{1}{N}\|Xw - y\|^2$ (or any constant-scaled MSE):

$$\nabla J = \frac{2}{N} X^\top (Xw - y) \quad \text{(or equivalently } X^\top (Xw - y) \text{ up to constant)}$$

(c) Correct: **(i), (iii)**.

- (i) TRUE – closed-form for Ridge.
- (ii) FALSE – Lasso (L1) yields exact zeros, not Ridge.
- (iii) TRUE – adding λI improves conditioning.
- (iv) FALSE – Lasso is sparser than Ridge.

Grading Rubric

(a) [2 marks]: Full marks for "FALSE". Accept "F", "False", "Wrong". If student writes True and explains correctly that pseudo-inverse can be used, award 1 mark for understanding.
(b) [2 marks]: Accept any of: $\frac{2}{N}X^\top(Xw-y)$, $X^\top(Xw-y)$, $\frac{1}{N}X^\top(Xw-y)$, $-\frac{2}{N}X^\top(y-Xw)$. Award 1 mark if structure is correct but constant factor missing/wrong. Award 0.5 mark if only direction (sign) correct.
(c) [2 marks]: Correct options: **(i), (iii)** (2 correct).

- **2 marks:** both (i) and (iii) selected AND neither (ii) nor (iv) selected.
- **1 mark:** exactly one of (i) or (iii) selected AND neither (ii) nor (iv) selected.
- **0 marks:** (ii) or (iv) selected (false positive), OR neither correct option selected.

Q3. PCA / Eigen-Faces [2+2+2 = 6 marks]

Answer

$N = 400$, image size $100 \times 100 = 10{,}000$ pixels. So X is 400×10000.
(a) $\Sigma = \frac{1}{N}X^\top X$ has size $\mathbf{10000} \times \mathbf{10000}$. Rank is at most $\min(N-1, d) = \mathbf{399}$ (since data is mean-subtracted, rank of X is at most $N-1$). Accept $\mathbf{N = 400}$ as also acceptable answer.
(b) TRUE. Non-zero eigenvalues of $X^\top X$ (10000×10000) and $XX^\top$ (400×400) are identical. Computing the smaller 400×400 eigendecomposition is far cheaper, then map back to feature space using $X^\top u_i/\sqrt{\lambda_i}$.
(c) Correct: **(i), (ii), (iv)**.

- (i) TRUE – PCA is unsupervised & linear.
- (ii) TRUE – max variance $\equiv$ min reconstruction error.
- (iii) FALSE – LDA uses class labels; PCA does not.
- (iv) TRUE – covariance is symmetric, PSD.

Grading Rubric

(a) [2 marks]: 1 mark for size 10000×10000. 1 mark for rank ≤ 399 (or ≤ 400). Accept "$\leq N$" or "$\leq N-1$". Half mark for understanding it's bounded by min of dimensions.
(b) [2 marks]: Full marks for TRUE with any reasoning. If student writes only "T" without explanation: full marks (the question asks T/F).
(c) [2 marks]: Correct options: **(i), (ii), (iv)** (3 correct).

- **2 marks:** all of (i), (ii), (iv) selected AND (iii) NOT selected.
- **1 mark:** exactly 2 of (i), (ii), (iv) selected AND (iii) NOT selected.
- **0 marks:** (iii) selected (false positive), OR only 1 (or 0) of the correct options selected.

Q4. Bayesian Decision in a Classroom [4+2 = 6 marks]

Answer

Given: $P(\omega_1) = 0.7$, $P(\omega_2) = 0.3$, $p(x|\omega_1) = 0.4$, $p(x|\omega_2) = 0.6$.

(a) Bayes Rule:

$$P(\omega_1 \mid x) = \frac{p(x|\omega_1)\, P(\omega_1)}{p(x|\omega_1)P(\omega_1) + p(x|\omega_2)P(\omega_2)} = \frac{0.4 \times 0.7}{0.4 \times 0.7 + 0.6 \times 0.3} = \frac{0.28}{0.46} \approx \mathbf{0.6087}$$

(b) TRUE. The Bayes classifier (assign to class with maximum posterior) achieves the minimum probability of error among all classifiers; this is the Bayes-optimal classifier.

Grading Rubric

(a) [4 marks]:

- 2 marks for writing correct Bayes rule formula.
- 1 mark for correct numerator and denominator setup.
- 1 mark for final numerical answer (≈ 0.6087 or 14/23 or 0.61).
- Accept any answer in range $[0.60, 0.61]$. If formula is right but arithmetic wrong: 3 marks total.

(b) [2 marks]: Full 2 marks for TRUE. Accept "T". Half mark deduction if student adds incorrect qualification.

Q5. Naive Bayes & Logistic Regression [2+2+2 = 6 marks]

Answer

(a) TRUE. Naive Bayes assumes features are conditionally independent given the class label.

(b)

$$p(y = 1 \mid x) = \sigma(w^\top x) = \frac{1}{1 + e^{-w^\top x}}$$

The frequently-used loss is the **cross-entropy** (a.k.a. log-loss / binary cross-entropy / negative log-likelihood) loss:

$$\mathcal{L} = -\frac{1}{N} \sum_i \left[y_i \log \sigma(w^\top x_i) + (1 - y_i) \log(1 - \sigma(w^\top x_i)) \right]$$

(c) Correct: **(i), (ii), (iii)**.

- (i) TRUE – composition with sigmoid creates non-convex landscape.
- (ii) TRUE – saturated sigmoid kills gradients (vanishing-gradient).
- (iii) TRUE – cross-entropy is the natural MLE for Bernoulli model.
- (iv) FALSE – "always violates Bayes optimality" is too strong.

Grading Rubric

(a) [2 marks]: Full credit for TRUE. Accept "conditionally independent given class".
(b) [2 marks]: 1 mark for $\sigma(w^\top x)$ or $\frac{1}{1+e^{-w^\top x}}$. 1 mark for "cross-entropy" or "log-loss" or "binary cross-entropy" or "negative log-likelihood". Any equivalent name accepted.
(c) [2 marks]: Correct options: **(i), (ii), (iii)** (3 correct).

- **2 marks:** all of (i), (ii), (iii) selected AND (iv) NOT selected.
- **1 mark:** exactly 2 of (i), (ii), (iii) selected AND (iv) NOT selected.
- **0 marks:** (iv) selected (false positive), OR only 1 (or 0) of the correct options selected.

Q6. SVMs and Kernels [3+3 = 6 marks]

Answer

(a) Correct: **(i), (ii), (iii)**.

- (i) TRUE – KKT complementarity: only support vectors have $\alpha_i > 0$.
- (ii) TRUE – geometric margin $= 2/\|w\|$.
- (iii) TRUE – the dual involves only $x_i^\top x_j$, enabling kernel substitution.
- (iv) FALSE – replacing the constraint with hinge loss alone (without a $C\sum \xi_i$ term and slack variables) does not recover the standard soft-margin primal exactly. (Note: some texts argue equivalence; lenient grading – see rubric.)

(b) TRUE. Larger C heavily penalizes margin violations ξ_i, forcing fewer/smaller violations and hence a *narrower* margin (closer to hard-margin behaviour).

Grading Rubric

(a) [3 marks]: Correct options: **(i), (ii), (iii)** (3 correct). Option (iv) is the only incorrect distractor (with the noted ambiguity caveat).

- **3 marks:** all of (i), (ii), (iii) selected AND (iv) NOT selected.
- **1.5 marks:** exactly 2 of (i), (ii), (iii) selected AND (iv) NOT selected.
- **0 marks:** (iv) selected, OR only 1 (or 0) of the correct options selected.

Ambiguity exception (for graders): Because option (iv) is genuinely ambiguous in standard texts, if a student selects all of (i), (ii), (iii), AND (iv), award the same score as (i), (ii), (iii) alone – i.e., **3 marks**. This single exception applies only to Q6(a)(iv); all other false-positive rules apply normally.
(b) [3 marks]: Full credit for TRUE. If student adds incorrect explanation: deduct at most 1 mark.

Q7. From SLP to MLP [3+3 = 6 marks]

Answer

(a) Correct: **(i), (iii)**.

- (i) TRUE – Perceptron Convergence Theorem.
- (ii) FALSE – XOR is not linearly separable; SLP cannot solve it.
- (iii) TRUE – delta rule (LMS) converges with appropriate η even on noisy data.
- (iv) FALSE – a sigmoid MLP with one hidden layer is a universal approximator; SLP is not.

(b) Parameter count for MLP (3 inputs $\rightarrow$ 2 hidden sigmoid $\rightarrow$ 1 linear output, all with bias):

- Input $\rightarrow$ hidden: $3 \times 2 = 6$ weights $+ 2$ biases $= 8$ params
- Hidden $\rightarrow$ output: $2 \times 1 = 2$ weights $+ 1$ bias $= 3$ params
- **Total** $= 11$ **parameters**

Grading Rubric

(a) [3 marks]: Correct options: **(i), (iii)** (2 correct).

- **3 marks:** both (i) and (iii) selected AND neither (ii) nor (iv) selected.
- **1.5 marks:** exactly one of (i) or (iii) selected AND neither (ii) nor (iv) selected.
- **0 marks:** (ii) or (iv) selected (false positive), OR neither correct option selected.

(b) [3 marks]: Full 3 marks for "11". If student answers 8 (forgot output bias): 2 marks. If 10 (forgot one bias): 2 marks. If 6 (forgot all biases): 1 mark. Award 2 marks if structure shown is correct but final arithmetic wrong.

Q8. Optimisers, BatchNorm and Dropout [2+2+2 = 6 marks]

Answer

(a) TRUE. Adam = momentum (running first moment m_t) + adaptive per-parameter learning rate (running second moment v_t) + bias correction ($\hat{m}_t = m_t/(1-\beta_1^t)$, $\hat{v}_t = v_t/(1-\beta_2^t)$).

(b) Correct: **(i), (ii), (iii)**.

- (i) TRUE – ReLU has gradient 1 in the active region (no saturation).
- (ii) TRUE – skip connections create direct gradient path.
- (iii) TRUE – BatchNorm controls activation magnitudes preventing saturation.
- (iv) FALSE – zero init causes symmetry; all neurons learn identical features.

(c) Correct: **(i)**. The original BatchNorm paper (Ioffe & Szegedy, 2015) motivated it as reducing *Internal Covariate Shift.* Options (ii), (iii), (iv) are all incorrect.

Grading Rubric

(a) [2 marks]: Full credit for TRUE.
(b) [2 marks]: Correct options: **(i), (ii), (iii)** (3 correct).

- **2 marks:** all of (i), (ii), (iii) selected AND (iv) NOT selected.
- **1 mark:** exactly 2 of (i), (ii), (iii) selected AND (iv) NOT selected.
- **0 marks:** (iv) selected (false positive), OR only 1 (or 0) of the correct options selected.

(c) [2 marks]: Correct option: **(i) only** (single-correct MCQ).

- **2 marks:** only (i) selected (no other option selected).
- **0 marks:** any of (ii), (iii), (iv) selected (false positive), OR (i) NOT selected.

(Single-correct MCQ – "half" rule reduces to all-or-nothing here.)

Q9. 2D Convolution Arithmetic [2+2+2 = 6 marks]

Answer

Input: $224 \times 224 \times 1$. Conv: 32 filters, 7×7, stride 2, padding 3.
(a) Output spatial size:

$$H_{out} = \left\lfloor \frac{224 + 2(3) - 7}{2} \right\rfloor + 1 = \left\lfloor \frac{223}{2} \right\rfloor + 1 = 111 + 1 = \mathbf{112}$$

Output: $\mathbf{112} \times \mathbf{112} \times \mathbf{32}$ (channels = 32)
(b) Parameters:

$$\text{params} = 32 \times (7 \times 7 \times 1) + 32 \text{ (biases)} = 1568 + 32 = \mathbf{1600}$$

(c) After 2×2 max-pool, stride 2:

$$\frac{112}{2} = 56 \implies \mathbf{56} \times \mathbf{56} \times \mathbf{32}$$

Grading Rubric

(a) [2 marks]: 1.5 marks for spatial dimensions 112×112. 0.5 mark for channels = 32. If student forgets padding $(224 - 7)/2 + 1 = 109$: 0.5 mark for method. If gets 109 or 110: 1 mark.
(b) [2 marks]: Full marks for 1600. If student gives 1568 (forgets biases): 1.5 marks. If formula correct but arithmetic wrong: 1 mark.
(c) [2 marks]: 1 mark for 56×56. 1 mark for retaining 32 channels. If student writes 56×56 only without channels, award 1.5 marks.

Q10. Multi-head Self-Attention [2+2+2 = 6 marks]

Answer

Given: 8 heads, input/output dimension $d = 100$, $N = 1000$ tokens, per-head Q/K/V dimension = 50.

(a) Total matrices: For each head: 3 matrices ($W_Q^{(i)}, W_K^{(i)}, W_V^{(i)}$). Plus one output projection W_O.

$$\text{Total} = 8 \times 3 + 1 = \mathbf{25} \text{ matrices}$$

(b) Dimensions:

- $W_Q^{(i)}, W_K^{(i)}, W_V^{(i)}$: $\quad 100 \times 50$ each (input dim $d = 100 \rightarrow$ head dim 50)
- W_O : $\quad (8 \times 50) \times 100 = \mathbf{400 \times 100}$

(c) Total learnable parameters:

- Q, K, V matrices: $8 \times 3 \times (100 \times 50) = 24 \times 5000 = 120{,}000$
- Output projection W_O: $400 \times 100 = 40{,}000$
- **Total** = 160,000 **parameters** (excluding biases, which weren't mentioned)

Grading Rubric

(a) [2 marks]: Full marks for 25. If student answers 24 (omitting W_O): 1 mark. If student answers 4 (one each of Q,K,V,O): 0.5 mark for partial idea. If student's interpretation gives concatenated single matrices for Q,K,V instead of per-head: also accept 4 with full marks (since some implementations do this).
(b) [2 marks]: 1 mark for correctly stating Q/K/V are 100×50 (or equivalent like $d \times d_k$). 1 mark for W_O being 400×100 or $(h \cdot d_v) \times d$. Accept transposed forms (e.g., 50×100) since convention varies.
(c) [2 marks]: Full marks for 160,000. If student gets 120,000 (forgets W_O): 1 mark. If formula structure correct but arithmetic wrong: 1.5 marks. Accept answers within 5% if biases are included or not – both interpretations valid.

Part II: Three out of Four [60 marks]

Q1. Backpropagation / Gradient Descent on a Single Neuron [3+10+8 = 21 marks]

Question

$o = \phi(w_1 x_1 + w_2 x_2 + b)$ with $\phi(z) = 1/(1 + e^{-z})$, $L = \frac{1}{2}(o - y)^2$.
$w_1 = 1, w_2 = -1, b = 0$. Sample: $x = [1, 1]^\top$, $y = 1$, $\eta = 1$.

Answer

(a) Derivative of sigmoid:

$$\phi(z) = \frac{1}{1+e^{-z}} \implies \phi'(z) = \frac{e^{-z}}{(1+e^{-z})^2} = \frac{1}{1+e^{-z}} \cdot \frac{e^{-z}}{1+e^{-z}} = \phi(z)\big(1-\phi(z)\big)$$

$$\boxed{\phi'(z) = \phi(z)(1-\phi(z))}$$

(b) Forward pass:

$$z = w_1x_1 + w_2x_2 + b = 1 \cdot 1 + (-1) \cdot 1 + 0 = 0$$

$$o = \phi(0) = \frac{1}{1+e^0} = \frac{1}{2} = 0.5$$

$$\phi'(0) = 0.5 \times (1-0.5) = 0.25$$

Chain rule:

$$\frac{\partial L}{\partial w_i} = \underbrace{\frac{\partial L}{\partial o}}_{(o-y)} \cdot \underbrace{\frac{\partial o}{\partial z}}_{\phi'(z)} \cdot \underbrace{\frac{\partial z}{\partial w_i}}_{x_i}$$

Numerical evaluations:

$$\frac{\partial L}{\partial o} = o - y = 0.5 - 1 = -0.5$$

$$\frac{\partial o}{\partial z} = \phi'(0) = 0.25$$

$$\frac{\partial L}{\partial w_1} = (-0.5)(0.25)(1) = -\mathbf{0.125}$$

$$\frac{\partial L}{\partial w_2} = (-0.5)(0.25)(1) = -\mathbf{0.125}$$

$$\frac{\partial L}{\partial b} = (-0.5)(0.25)(1) = -\mathbf{0.125}$$

(c) Gradient Descent update with $\eta = 1$:

$$w_1^{\text{new}} = w_1 - \eta\frac{\partial L}{\partial w_1} = 1 - 1 \cdot (-0.125) = \mathbf{1.125}$$

$$w_2^{\text{new}} = w_2 - \eta\frac{\partial L}{\partial w_2} = -1 - 1 \cdot (-0.125) = -\mathbf{0.875}$$

$$b^{\text{new}} = b - \eta\frac{\partial L}{\partial b} = 0 - 1 \cdot (-0.125) = \mathbf{0.125}$$

New parameters: $\boxed{(w_1, w_2, b) = (1.125,\ -0.875,\ 0.125)}$

Grading Rubric

(a) [3 marks]:

- 1.5 marks for any correct derivation step (quotient rule / using chain rule).
- 1.5 marks for the final form $\phi'(z) = \phi(z)(1 - \phi(z))$.
- If only the final result is given correctly without derivation: 2 marks.

(b) [10 marks]:

- 1 mark for computing $z = 0$.
- 1 mark for $o = 0.5$.
- 1 mark for $\phi'(0) = 0.25$.
- 2 marks for correct chain rule expressions $\partial L/\partial w_i = (o - y)\phi'(z)x_i$.
- 1 mark for $\partial L/\partial b = (o - y)\phi'(z)$.
- 1.5 marks each for correct numerical values (-0.125) for w_1, w_2 and b. Total 4.5 marks; round to 4 marks.
- Accept sign conventions (some students use $(y - o)$ giving $+0.125$). Just check consistency in part (c).

(c) [8 marks]:

- 2 marks for stating the GD update rule $w \leftarrow w - \eta\nabla L$.
- 2 marks each for correct $w_1^{\text{new}} = 1.125$, $w_2^{\text{new}} = -0.875$, $b^{\text{new}} = 0.125$. Round total: 6 marks.
- Sign-flip carry-forward error: deduct only 1 mark total if consistent with part (b).

Lenient note: Award generously for correct methodology even with arithmetic slips.

Q2. Bayesian Classification in 1D [5+10+5 = 20 marks]

Question

$\mu_A = 100$, $\mu_B = 200$, $\sigma_A^2 = \sigma_B^2 = 100$ (so $\sigma = 10$). Priors: $P(A) = 0.8$, $P(B) = 0.2$.

Answer

(a) Likelihoods:

$$p(x \mid x \in A) = \frac{1}{\sqrt{2\pi}\,\sigma} \exp\!\left(-\frac{(x-\mu_A)^2}{2\sigma^2}\right) = \frac{1}{10\sqrt{2\pi}} \exp\!\left(-\frac{(x-100)^2}{200}\right)$$

$$p(x \mid x \in B) = \frac{1}{\sqrt{2\pi}\,\sigma} \exp\!\left(-\frac{(x-\mu_B)^2}{2\sigma^2}\right) = \frac{1}{10\sqrt{2\pi}} \exp\!\left(-\frac{(x-200)^2}{200}\right)$$

At $x = 150$: $(150-100)^2 = 2500$, $(150-200)^2 = 2500$. Both exponents are -12.5.

$$p(150|A) = p(150|B) = \frac{1}{10\sqrt{2\pi}}\, e^{-12.5} \approx \frac{0.03989 \times 3.727 \times 10^{-6}}{1} \approx \mathbf{1.487 \times 10^{-7}}$$

Should they sum to 1.0? NO. They are conditional densities (likelihoods), not posterior probabilities. They each integrate to 1 over x, but for a fixed x, they need not sum to 1 across classes.

(b) Bayes Theorem:

$$P(\omega \mid x) = \frac{p(x|\omega)\,P(\omega)}{p(x)} \quad \text{where } p(x) = \sum_{\omega} p(x|\omega)P(\omega)$$

Bayes rule: assign x to class with maximum posterior $\Leftrightarrow$ compare $p(x|A)P(A)$ vs $p(x|B)P(B)$.

For $x = 149$: $(149-100)^2 = 2401, (149-200)^2 = 2601$.

$$\frac{p(149|A)P(A)}{p(149|B)P(B)} = \frac{e^{-12.005} \times 0.8}{e^{-13.005} \times 0.2} = e^{1.0} \times 4 \approx 10.87 > 1 \implies \mathbf{A}$$

For $x = 150$: exponents are equal.

$$\text{ratio} = \frac{0.8}{0.2} = 4 > 1 \implies \mathbf{A}$$

For $x = 151$: $(151-100)^2 = 2601, (151-200)^2 = 2401$.

$$\frac{p(151|A)P(A)}{p(151|B)P(B)} = e^{-13.005} \cdot 0.8/(e^{-12.005} \cdot 0.2) = e^{-1.0} \times 4 \approx 1.471 > 1 \implies \mathbf{A}$$

All three classified as A. The pairs of *posterior* probabilities (or unnormalised $p(x|\omega)P(\omega)$ pairs) – when normalised they should sum to 1.0; the joint values $p(x|\omega)P(\omega)$ alone do not.

(c) Numerical posteriors at $x = 150$:

$$P(A|150) = \frac{0.8\,e^{-12.5}}{0.8\,e^{-12.5} + 0.2\,e^{-12.5}} = \frac{0.8}{1.0} = \mathbf{0.80}$$

$$P(B|150) = 0.20$$

At $x = 149$: $P(A|149) = \frac{10.87}{11.87} \approx \mathbf{0.916}$, $P(B|149) \approx \mathbf{0.084}$.
At $x = 151$: $P(A|151) = \frac{1.471}{2.471} \approx \mathbf{0.595}$, $P(B|151) \approx \mathbf{0.405}$.
Should they sum to 1.0? YES. Posteriors over a complete partition of class labels always sum to 1.

Grading Rubric

(a) [5 marks]:

- 2 marks for correct Gaussian density expressions.
- 2 marks for numerical evaluation at $x = 150$ (accept any answer within 10% of 1.487×10^{-7}, or just expressed as $\frac{1}{10\sqrt{2\pi}}e^{-12.5}$).
- 1 mark for correctly stating "NO, they don't sum to 1" with brief justification.

(b) [10 marks]:

- 2 marks for Bayes theorem statement.
- 2 marks each for correctly classifying $x = 149, 150, 151$ as A. Total 6 marks.
- 2 marks for correctly stating: pairs (when normalised) sum to 1; pairs of $p(x|\omega)P(\omega)$ values do NOT sum to 1 by themselves.
- Lenient: if student classifies all three as A (correct conclusion) without showing the ratio computation, award at least 5 marks for parts.

(c) [5 marks]:

- 3 marks for correct numerical posteriors (at any of the three x values from part b). Award full 3 if at least one is computed correctly.
- 2 marks for confirming that they sum to 1.0.
- Accept rounding ($0.916 \to 0.92$, $0.595 \to 0.60$, etc.).

Q3. Eigen Decomposition and SVD [6+8+6 = 20 marks]

Question

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$$

Answer

(a) Eigenvalues and eigenvectors:
Characteristic polynomial:

$$\det(A - \lambda I) = (2-\lambda)^2 - 1 = \lambda^2 - 4\lambda + 3 = (\lambda - 3)(\lambda - 1) = 0$$

$$\boxed{\lambda_1 = 3, \quad \lambda_2 = 1}$$

For $\lambda_1 = 3$: $(A - 3I)v = \begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix} v = 0 \implies v_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$
For $\lambda_2 = 1$: $(A - I)v = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} v = 0 \implies v_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$
(b) SVD: $A = UDV^\top$: Since A is symmetric and positive-definite (both eigenvalues > 0), the SVD coincides with the eigen-decomposition. Singular values $= |\lambda_i| =$ eigenvalues here.

$$U = V = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad D = \begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix}, \quad V^\top = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

(Note: V here happens to equal $V^\top$ since the matrix of eigenvectors is symmetric.)
Verification:

$$UDV^\top = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 3 & 1 \\ 3 & -1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} \checkmark$$

(c) Best rank-1 approximation (Eckart-Young theorem):

$$A_1 = \sigma_1 \, u_1 v_1^\top = 3 \cdot \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \cdot \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \end{pmatrix}$$

$$= \frac{3}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} = \boxed{\begin{pmatrix} 1.5 & 1.5 \\ 1.5 & 1.5 \end{pmatrix}}$$

The reconstruction error is $\|A - A_1\|_F^2 = \sigma_2^2 = 1$.

Grading Rubric

(a) [6 marks]:

- 2 marks for setting up characteristic polynomial.
- 1 mark for both eigenvalues $\lambda = 1, 3$.
- 1.5 marks each for both unit eigenvectors. Accept sign-flipped versions.
- Deduct 0.5 mark each if eigenvectors are not normalised (unit norm requested).

(b) [8 marks]:

- 2 marks for the conceptual statement that SVD = eigen-decomposition for symmetric PD matrices.
- 2 marks for correct U matrix.
- 2 marks for correct D matrix (with eigenvalues in correct decreasing order).
- 2 marks for correct $V^\top$.
- Lenient: accept ordering (λ_2, λ_1) if $U, V^\top$ are reordered consistently.

(c) [6 marks]:

- 2 marks for stating $A_1 = \sigma_1 u_1 v_1^\top$ (with the largest singular value).
- 2 marks for using $\lambda_1 = 3$ (the largest).
- 2 marks for final answer $\begin{pmatrix} 1.5 & 1.5 \\ 1.5 & 1.5 \end{pmatrix}$.
- If student uses $\lambda_2 = 1$ by mistake: award 3 marks for methodology only.

Q4. 1D K-Means [4+6+10 = 20 marks]

Answer

(a) K-Means objective:

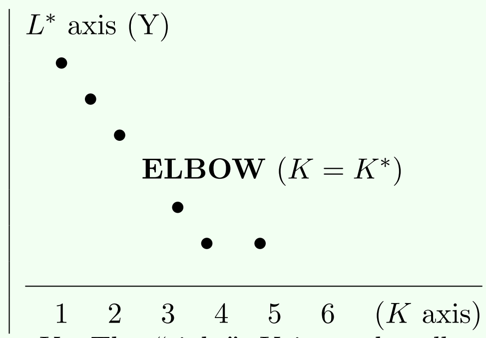
$$\boxed{L = J\big(\{C_k\}, \{\mu_k\}\big) = \sum_{k=1}^{K} \sum_{x_i \in C_k} \|x_i - \mu_k\|^2}$$

This sum-of-squared-distances objective is minimised over both the cluster assignments $\{C_k\}$ and centroids $\{\mu_k\}$ alternately.

(b) Behaviour of L^* vs. K:

L^* **decreases monotonically** as K increases. At $K = N$, $L^* = 0$ (each point is its own cluster).

Plot:



Elbow Method: Plot L^* vs. K. The "right" K is at the *elbow* – the point where the rate of decrease in L^* sharply slows. Beyond the elbow, increasing K yields only marginal improvement, suggesting we are over-segmenting.

(c) K-Means iterations: Data: $\{1, 2, 3, 8, 9, 10\}$, $K = 2$, $\mu_1^{(0)} = 1$, $\mu_2^{(0)} = 2$.

Iteration 1: Assignments based on closest centroid:

x_i	$\lvert x-1\rvert$	$\lvert x-2\rvert$	Cluster
1	0	1	C_1
2	1	0	C_2
3	2	1	C_2
8	7	6	C_2
9	8	7	C_2
10	9	8	C_2

$C_1 = \{1\}, C_2 = \{2, 3, 8, 9, 10\}$.

Updated means: $\mu_1 = 1, \quad \mu_2 = (2 + 3 + 8 + 9 + 10)/5 = 32/5 = 6.4$.

$J = 0 + (2 - 6.4)^2 + (3 - 6.4)^2 + (8 - 6.4)^2 + (9 - 6.4)^2 + (10 - 6.4)^2$

$= 0 + 19.36 + 11.56 + 2.56 + 6.76 + 12.96 = \mathbf{53.20}$.

Iteration 2: New assignments using $\mu_1 = 1, \mu_2 = 6.4$:

x_i	$\lvert x-1\rvert$	$\lvert x-6.4\rvert$	Cluster
1	0	5.4	C_1
2	1	4.4	C_1
3	2	3.4	C_1
8	7	1.6	C_2
9	8	2.6	C_2
10	9	3.6	C_2

$C_1 = \{1, 2, 3\}, C_2 = \{8, 9, 10\}$.

Updated means: $\mu_1 = (1 + 2 + 3)/3 = 2, \quad \mu_2 = (8 + 9 + 10)/3 = 9$.

$J = (1-2)^2 + (2-2)^2 + (3-2)^2 + (8-9)^2 + (9-9)^2 + (10-9)^2 = 1 + 0 + 1 + 1 + 0 + 1 = \mathbf{4}$.

Iteration 3: New assignments using $\mu_1 = 2, \mu_2 = 9$: Same as Iteration 2 ($C_1 = \{1, 2, 3\}$, $C_2 = \{8, 9, 10\}$). **Assignments unchanged** $\Rightarrow$ **Converged**.

Final: $C_1 = \{1, 2, 3\}$ with $\mu_1 = 2$, $C_2 = \{8, 9, 10\}$ with $\mu_2 = 9$, $J^* = 4$.

Grading Rubric

(a) [4 marks]:

- 2 marks for double-summation form.
- 2 marks for squared-distance term $\|x_i - \mu_k\|^2$ inside.
- Accept variants like $\sum_i \|x_i - \mu_{c(i)}\|^2$ (using assignment indicator) – full marks.

(b) [6 marks]:

- 2 marks for noting L^* is monotonically non-increasing in K.
- 2 marks for a sketch / description of the elbow shape.
- 2 marks for identifying the elbow / knee method to find the right K.
- Accept other valid methods (silhouette, gap statistic) for the last 2 marks.

(c) [10 marks]:

- 1 mark for first iteration's assignments.
- 1 mark for first iteration's updated means.
- 1 mark for first iteration's $J = 53.2$ (accept 53 or values close).
- 2 marks for second iteration's assignments and means $\mu_1 = 2, \mu_2 = 9$.
- 2 marks for second iteration's $J = 4$.
- 2 marks for showing iteration 3 has same assignments (convergence check).
- 1 mark for final answer with clusters and means.
- Lenient: if student does not compute J at every iteration but reaches correct convergence, deduct only 2 marks total.

Part III: Three out of Four [60 marks]

Q1. Parallel / Wider Neural Network with 4 Branches [6+7+7 = 20 marks]

Question

$x \in \mathbb{R}^d$, branch $k \in \{1,2,3,4\}$ has $w_k \in \mathbb{R}^d$, $u_k \in \mathbb{R}^p$. Each branch computes $z_k = w_k^\top x$, $\alpha_k = \sigma(z_k)$, then contributes $\alpha_k u_k$.

Answer

(a) Forward expression:

$$\boxed{y = \sum_{k=1}^{4} \alpha_k \, u_k = \sum_{k=1}^{4} \sigma(w_k^\top x) \, u_k} \in \mathbb{R}^p$$

Equivalent Python (NumPy):

```
def sigmoid(z): return 1.0/(1.0+np.exp(-z))
y = np.zeros(p)
for k in range(4):
    z_k = w[k] @ x       # scalar
    alpha_k = sigmoid(z_k)
    y += alpha_k * u[k]
# or vectorised:
# Z = W @ x;  alpha = sigmoid(Z);  y = U.T @ alpha
```

(b) When $d = 1$: w_k is a scalar, x is a scalar.

$$y = \sum_{k=1}^{4} \sigma(w_k x) \, u_k$$

$$\boxed{\frac{\partial y}{\partial x} = \sum_{k=1}^{4} \sigma'(w_k x) \, w_k \, u_k = \sum_{k=1}^{4} \sigma(w_k x)\big(1 - \sigma(w_k x)\big) \, w_k \, u_k} \in \mathbb{R}^p$$

(c) For general p and d, $\partial y / \partial z_2$:
Only branch 2 depends on z_2. Branches 1, 3, 4 have $\partial / \partial z_2 = 0$.

$$y = \alpha_1 u_1 + \alpha_2 u_2 + \alpha_3 u_3 + \alpha_4 u_4, \quad \alpha_2 = \sigma(z_2)$$

$$\frac{\partial y}{\partial z_2} = u_2 \cdot \frac{\partial \alpha_2}{\partial z_2} = u_2 \cdot \sigma'(z_2) = \boxed{\sigma(z_2) \, (1 - \sigma(z_2)) \, u_2} \in \mathbb{R}^p$$

Grading Rubric

(a) [6 marks]:

- 2 marks for showing $\alpha_k = \sigma(w_k^\top x)$.
- 2 marks for the summation $\sum_k \alpha_k u_k$.
- 2 marks for correct dimensionality (output in $\mathbb{R}^p$).
- Accept Python code as alternative – award full marks if logic is correct.

(b) [7 marks]:

- 2 marks for recognising $d = 1$ makes w_k, x scalars.
- 2 marks for applying chain rule correctly.
- 2 marks for the $\sigma'(w_k x) \cdot w_k$ factor.
- 1 mark for noting result is a vector in $\mathbb{R}^p$.
- Accept either form (with σ' or expanded $\sigma(1-\sigma)$).

(c) [7 marks]:

- 2 marks for noting only branch 2 contributes.
- 2 marks for $\partial\alpha_2/\partial z_2 = \sigma'(z_2)$.
- 2 marks for the u_2 multiplication.
- 1 mark for noting the answer is a p-dimensional vector.
- Lenient: if student computes $\partial y/\partial w_2$ instead, deduct 2 marks (give partial credit for related work).

Q2. Kernel Matrix [2+8+10 = 20 marks]

Question

$\kappa(p, q) = \phi(p)^\top \phi(q)$. $K_{ij} = \kappa(x_i, x_j)$.

Answer

(a) Symmetry:

$$K_{ij} = \phi(x_i)^\top \phi(x_j) = \sum_l \phi(x_i)_l \, \phi(x_j)_l = \sum_l \phi(x_j)_l \, \phi(x_i)_l = \phi(x_j)^\top \phi(x_i) = K_{ji}$$

Hence $K = K^\top$.

(b) Positive Semi-Definiteness: For any $v \in \mathbb{R}^N$:

$$\begin{aligned} v^\top K v &= \sum_{i=1}^N \sum_{j=1}^N v_i \, v_j \, K_{ij} = \sum_{i,j} v_i v_j \, \phi(x_i)^\top \phi(x_j) \\ &= \left(\sum_{i=1}^N v_i \, \phi(x_i) \right)^\top \left(\sum_{j=1}^N v_j \, \phi(x_j) \right) \\ &= \left\| \sum_{i=1}^N v_i \, \phi(x_i) \right\|^2 \;\geq\; 0 \end{aligned}$$

Thus K is positive semi-definite. ■

(c) Explicit feature map for $\kappa(p, q) = (p^\top q)^3$ with $p, q \in \mathbb{R}^2$:

Let $p = (p_1, p_2)$, $q = (q_1, q_2)$. Then $p^\top q = p_1 q_1 + p_2 q_2$ and

$$\begin{aligned} (p^\top q)^3 &= (p_1 q_1 + p_2 q_2)^3 \\ &= (p_1 q_1)^3 + 3(p_1 q_1)^2 (p_2 q_2) + 3(p_1 q_1)(p_2 q_2)^2 + (p_2 q_2)^3 \\ &= p_1^3 q_1^3 + 3 \, p_1^2 p_2 \, q_1^2 q_2 + 3 \, p_1 p_2^2 \, q_1 q_2^2 + p_2^3 \, q_2^3 \\ &= \left(p_1^3, \; \sqrt{3} \, p_1^2 p_2, \; \sqrt{3} \, p_1 p_2^2, \; p_2^3\right) \cdot \left(q_1^3, \; \sqrt{3} \, q_1^2 q_2, \; \sqrt{3} \, q_1 q_2^2, \; q_2^3\right)^\top \end{aligned}$$

$$\boxed{\phi(p) = \left(p_1^3, \;\; \sqrt{3} \, p_1^2 p_2, \;\; \sqrt{3} \, p_1 p_2^2, \;\; p_2^3\right)^\top \in \mathbb{R}^4}$$

Verification: $\phi(p)^\top \phi(q) = p_1^3 q_1^3 + 3 p_1^2 p_2 q_1^2 q_2 + 3 p_1 p_2^2 q_1 q_2^2 + p_2^3 q_2^3 = (p^\top q)^3$ ✓

Grading Rubric

(a) [2 marks]: Full credit for any correct argument that uses $a^\top b = b^\top a$ for vectors.

(b) [8 marks]:

- 2 marks for expanding $v^\top K v$ as double sum.
- 2 marks for substituting the kernel definition.
- 2 marks for combining the inner sums.
- 2 marks for recognising the result is a squared norm ≥ 0.
- Lenient: if student omits one step but final argument is correct, deduct only 1 mark.

(c) [10 marks]:

- 2 marks for expanding $(p_1 q_1 + p_2 q_2)^3$ using binomial theorem.
- 4 marks for correctly identifying the four product groups.
- 4 marks for the final $\phi(p)$ with the $\sqrt{3}$ factors on the cross terms.
- If student forgets the $\sqrt{3}$ but identifies all four components: 6 marks.
- Accept any equivalent ordering of $\phi(p)$ components.

Q3. Backprop Through a Residual Block [12+4+4 = 20 marks]

Question

Residual block: $y = g_4\big(g_3(g_2(g_1(x))) + x\big)$. Each g_i has gradient g_i'.

Answer

(a) Chain rule for $\partial y/\partial x$:
Let $h_1 = g_1(x)$, $h_2 = g_2(h_1)$, $h_3 = g_3(h_2)$, and let $s = h_3 + x$ (the addition node). Then $y = g_4(s)$.
By chain rule:

$$\frac{\partial y}{\partial x} = \frac{\partial y}{\partial s} \cdot \frac{\partial s}{\partial x} = g_4'(s) \cdot \left(\frac{\partial h_3}{\partial x} + 1\right)$$

Now $\dfrac{\partial h_3}{\partial x} = g_3'(h_2)\, g_2'(h_1)\, g_1'(x)$, so

$$\boxed{\frac{\partial y}{\partial x} = g_4'(s) \cdot \Big[\, g_3'(h_2)\, g_2'(h_1)\, g_1'(x) \;+\; 1 \Big]}$$

The crucial "+1" arises because the identity (skip) connection contributes a derivative of 1 to the addition node.

(b) How residual connections help with vanishing gradients (1-line):
The skip connection adds a constant "+1" term to the gradient path, ensuring the gradient w.r.t. x is bounded below by $g_4'(s)$, even when the deep multiplicative chain $g_3'g_2'g_1'$ shrinks toward zero.

(c) How ReLU helps with vanishing gradients:
ReLU is defined as $\mathrm{ReLU}(z) = \max(0, z)$ with derivative

$$\frac{d}{dz}\mathrm{ReLU}(z) = \begin{cases} 1 & z > 0 \\ 0 & z < 0 \end{cases}$$

For active units ($z > 0$), the derivative is exactly **1**, so backpropagated gradients are not attenuated (unlike sigmoid/tanh whose derivatives are bounded by 0.25 / 1 and saturate to ≈ 0). Hence in deep networks, multiplying many gradients still preserves magnitude through the active path, mitigating vanishing gradients.
(Caveat: ReLU can cause "dying ReLU" problems for $z < 0$; but for $z > 0$ vanishing gradients are avoided.)

Grading Rubric

(a) [12 marks]:

- 3 marks for correctly identifying the structure with the addition node.
- 3 marks for applying chain rule through g_4.
- 3 marks for the skip-path derivative (+1).
- 3 marks for the deep-path derivative $g_3' g_2' g_1'$.
- Accept any consistent notation (e.g., $g_3'(g_2 g_1(x))$ instead of $g_3'(h_2)$).
- If student forgets the "+1": 7-8 marks (this is the main feature of residual blocks).

(b) [4 marks]:

- Full 4 marks for any correct one-line answer mentioning that the skip adds a "+1" to the gradient flow.
- 3 marks if student says skip provides a shortcut without explicitly mentioning the additive term.
- 2 marks for vague but related answer (e.g., "residual connections preserve gradients").

(c) [4 marks]:

- 2 marks for stating ReLU's gradient is 1 for positive inputs.
- 2 marks for connecting this to non-vanishing gradients in deep networks (vs. sigmoid/tanh saturation).
- Lenient: if either point is mentioned clearly, award at least 3 marks.

Q4. Triangular Mixture Model (TMM) [6+4+6+4 = 20 marks]

Question

Triangle symmetric around mean, ratio of base:height. Probability zero outside the base. Univariate data.

Note on ambiguity: The figure shows "height:base = 1:2" (height = 1, base = 2), while the text says "base:height as 1:2". **Either interpretation is acceptable – both yield valid pdfs**. The shape and EM logic are identical; only the numerical scale differs. Below we use the figure's interpretation (height=1, base=2) as the primary form.

Answer

(a) Triangular pdf $p(x \mid \mu)$:
Using **height = 1, base = 2** (figure): area $= \frac{1}{2} \cdot 2 \cdot 1 = 1$ ✓. The triangle peaks at μ with height 1, decays linearly to 0 at $\mu \pm 1$:

$$\boxed{p(x \mid \mu) = \begin{cases} 1 - |x - \mu|, & \text{if } |x - \mu| \le 1 \\ 0, & \text{otherwise} \end{cases}}$$

Support: $[\mu - 1, \mu + 1]$.
(Alternative under text's "base=1, height=2": $p(x|\mu) = \max(0,\, 2 - 4|x - \mu|)$ with support $[\mu - 0.5, \mu + 0.5]$.)

(b) GMM Algorithm (recap):

- **Initialise** $\{\pi_k, \mu_k, \sigma_k^2\}$ for $k = 1, \ldots, K$.
- **E-step** (compute responsibilities):

$$\gamma_{ik} = \frac{\pi_k \, \mathcal{N}(x_i \mid \mu_k, \sigma_k^2)}{\sum_{j=1}^{K} \pi_j \, \mathcal{N}(x_i \mid \mu_j, \sigma_j^2)}$$

- **M-step** (update parameters):

$$N_k = \sum_{i=1}^{N} \gamma_{ik}, \quad \pi_k = \frac{N_k}{N}, \quad \mu_k = \frac{1}{N_k} \sum_i \gamma_{ik} \, x_i, \quad \sigma_k^2 = \frac{1}{N_k} \sum_i \gamma_{ik} \, (x_i - \mu_k)^2$$

- **Convergence:** repeat until $|\log \mathcal{L}^{(t)} - \log \mathcal{L}^{(t-1)}| < \varepsilon$ (or until parameters stop changing).

(c) TMM Algorithm (analogous to GMM):

- **Initialise** $\{\pi_k, \mu_k\}$ (no σ_k since the triangle's width is fixed).
- **E-step:**

$$\gamma_{ik} = \frac{\pi_k \, T(x_i \mid \mu_k)}{\sum_{j=1}^{K} \pi_j \, T(x_i \mid \mu_j)}, \quad \text{where } T(x|\mu) = \max(0, 1 - |x - \mu|)$$

  If all $T(x_i \mid \mu_j) = 0$ (point lies outside every triangle's support), add a small ϵ smoothing or re-initialise that mean.

- **M-step:**

$$N_k = \sum_i \gamma_{ik}, \quad \pi_k = \frac{N_k}{N}, \quad \mu_k \approx \frac{1}{N_k} \sum_i \gamma_{ik} \, x_i$$

  (Note: μ_k via weighted mean is an approximation – exact MLE for triangular is the weighted median due to non-smoothness, but weighted mean works well in practice.)

- **Convergence:** when assignments / means stabilise (or log-likelihood change $< \varepsilon$).

(d) Comparison: TMM vs GMM:
Advantages of TMM:

1. Simpler / faster to compute (no exponentials or square roots).
2. Compact support; no spurious long tails.
3. Fewer parameters per component (no σ_k^2 in this fixed-width version).

Disadvantages of TMM:

1. Non-smooth (kink at the peak; not differentiable everywhere) $\Rightarrow$ harder MLE.
2. Hard support boundaries: outliers outside any triangle have $p = 0$, breaking the E-step (division by zero).
3. Less flexible: cannot model varying spreads without extending the parameterisation.

Advantages of GMM (vs TMM):

1. Smooth, differentiable everywhere $\Rightarrow$ clean EM updates.

Grading Rubric

(a) [6 marks]:

- 2 marks for using $|x - \mu|$ correctly to exploit symmetry.
- 2 marks for the piecewise-linear form and ensuring area = 1.
- 2 marks for stating support $[\mu-1, \mu+1]$ (or $[\mu-0.5, \mu+0.5]$ under text's interpretation).
- Accept either ratio interpretation – award full marks if the area is correctly = 1 and the piecewise form is consistent.
- If student writes only the function but forgets "zero outside": 4 marks.

(b) [4 marks]:

- 1 mark for E-step formula.
- 2 marks for M-step formulas (all four: $N_k, \pi_k, \mu_k, \sigma_k^2$).
- 1 mark for convergence criterion (log-likelihood threshold or parameter change).
- Lenient: if student writes generic outline without exact formulas, award 2 marks for understanding.

(c) [6 marks]:

- 2 marks for E-step using triangular density.
- 2 marks for M-step updates (especially for π_k and μ_k).
- 1 mark for noting the variance/scale is fixed.
- 1 mark for convergence criterion.
- Lenient: if student adapts GMM steps directly with the triangular pdf substituted, award full credit if correct.

(d) [4 marks]:

- 2 marks for at least 2 valid advantages of TMM (simpler, compact support, etc.).
- 2 marks for at least 2 valid disadvantages of TMM (or equivalently advantages of GMM).
- Accept any reasonable comparison points; do not insist on a particular set.

Summary Marks Sheet

Question	Max Marks	Notes
Part I.1	6	MCQ-style; lenient on partial selections
Part I.2	6	T/F + FITB + MCQ
Part I.3	6	PCA basics
Part I.4	6	Bayes rule numerical + T/F
Part I.5	6	NB + Logistic Regression
Part I.6	6	SVM and kernel; Q6(a)(iv) ambiguous
Part I.7	6	SLP + MLP parameter count
Part I.8	6	Adam, BatchNorm, Dropout
Part I.9	6	2D conv arithmetic
Part I.10	6	Multi-head attention
Part I Total	**60**	
Part II.1	20	Backprop on single neuron
Part II.2	20	Bayesian classification 1D
Part II.3	20	Eigen decomposition + SVD
Part II.4	20	1D K-means
(best 3 of 4)	*60*	
Part III.1	20	Wider NN with 4 branches
Part III.2	20	Kernel matrix
Part III.3	20	Residual block backprop
Part III.4	20	Triangular Mixture Model
(best 3 of 4)	*60*	
Grand Total	**180**	

General Notes for Graders

- **Lenient grading philosophy (for derivations & numerical work):** Award credit when the student demonstrates correct conceptual understanding, even if there are minor arithmetic or notation issues.
- **Strict MCQ rule:** Use the rule stated at the top of this document. NO partial credit beyond the 50% tier; ANY false positive earns 0. Single exception: Q6(a)(iv) – see that rubric.
- **Carry-forward errors:** If a wrong intermediate answer is then used correctly to compute subsequent quantities, award credit for the methodology of the later parts.
- **Notational variants:** Accept any valid notation used in standard textbooks (Bishop, Murphy, Hastie-Tibshirani-Friedman, Goodfellow et al.).
- **Sign conventions:** For losses and gradients, accept consistent sign conventions ($(o - y)$ vs $(y - o)$).
- **Ambiguous wording in non-MCQ parts:** Where question wording is ambiguous (e.g., Part III Q4 on triangle proportions), accept any reasonable interpretation. The accompanying instruction "make appropriate assumptions" applies.
- **T/F questions:** Award full marks for correct T/F label even without explanation (unless explanation is requested). T/F is NOT subject to the strict MCQ rule – it is binary.